

LAWANYA KHANDELWAL

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Objective

Seeking internships or co-ops in Electrical Engineering, ideally with a focus on digital systems, embedded design, or signal processing.

Education

Iowa State University – Ames, IA

Expected May 2028

Bachelor of Science in Electrical Engineering

Concurrent Degree: B.S. in Electrical Engineering (ISU) & B.E. in Electronics & Communication (BITS Pilani)

Birla Institute of Technology and Science – Pilani, India

2024 – 2026

Bachelor of Engineering (B.E.) in Electronics & Communication Engineering

GPA: 6.28/10.00

Select Academic Projects

FPGA-Based Digital Systems & Logic Circuit Design Aug 2025 – May 2026

Synthesized digital designs onto FPGA boards and built hardware prototypes using 74xx series TTL logic ICs on breadboards

Programmed MATLAB and Python scripts to compute Continuous-Time and Discrete-Time Fourier Transforms (CTFT/DTFT) and signal convolutions

Discrete Transistor Amplifiers & Circuit Analysis Aug 2026 – May 2026

Designed, biased, and assembled multi-stage BJT/MOSFET amplifiers, differential pairs, and active filter networks

Executed DC, AC sweep, and transient analyses in LTspice to optimize signal bandwidth, voltage gain, and Total Harmonic Distortion (THD)

Precision Analog Systems & Active Filter Design Aug 2026 – May 2026

Built and validated operational-amplifier circuits, including precision rectifiers, active band-pass filters, and waveform generators

Performed signal integrity checks using oscilloscopes and function generators to reconcile physical lab measurements with theoretical models

8086 Microprocessor Architecture & Peripheral Interfacing Aug 2026 – May 2026

Developed low-level 8086 assembly routines for data manipulation, string operations, and arithmetic processing

Integrated 8086 microprocessors to control stepper motors, matrix keypads, and LCDs

Relevant Skills

Hardware: Oscilloscopes, function generators, FPGA boards, 74xx TTL ICs, breadboard prototyping, BJT/MOSFET active filter networks

Software Applications and Languages: Verilog HDL, Xilinx Vivado, LTspice, MATLAB, Simulink, Cadence, AutoCAD, Assembly (ARM/x86), C, C++, Python, Java, SQL, HTML, Figma, MS Office

Leadership & Activities

Society for Medical and Aid Assistance (SMAC) – *Joint Co-Ordinator*

Spearheaded project planning and logistics for medical/first-aid facilities during annual cultural/technical festivals (Oasis/Atmos) and sports meets

Managed event logistics and directed an operational budget of \$20,000+

Women in Stem Engineering (WISE) – *Core Committee Member*

Youth Outreach & Inspiration and Career Development & Leadership programs

National Service Scheme (NSS) – *Core Committee Member*

Managed core operations for social service initiatives including blood donation drives, educational support programs, and book drives

Performing Arts & Cultural Clubs – *Member & Individual Performer*

MIME Performing Art Group: Performed UV acts, musicals, and stage dramas; assisted in video editing and production

Bharatanatyam: Completed Madhyama Purna certification for classical Indian dance

Shotokan Karate: Achieved Black Belt certification